

## SKILL 2 - PREPROCESSING REPORT

### One Piece Bounty Dataset

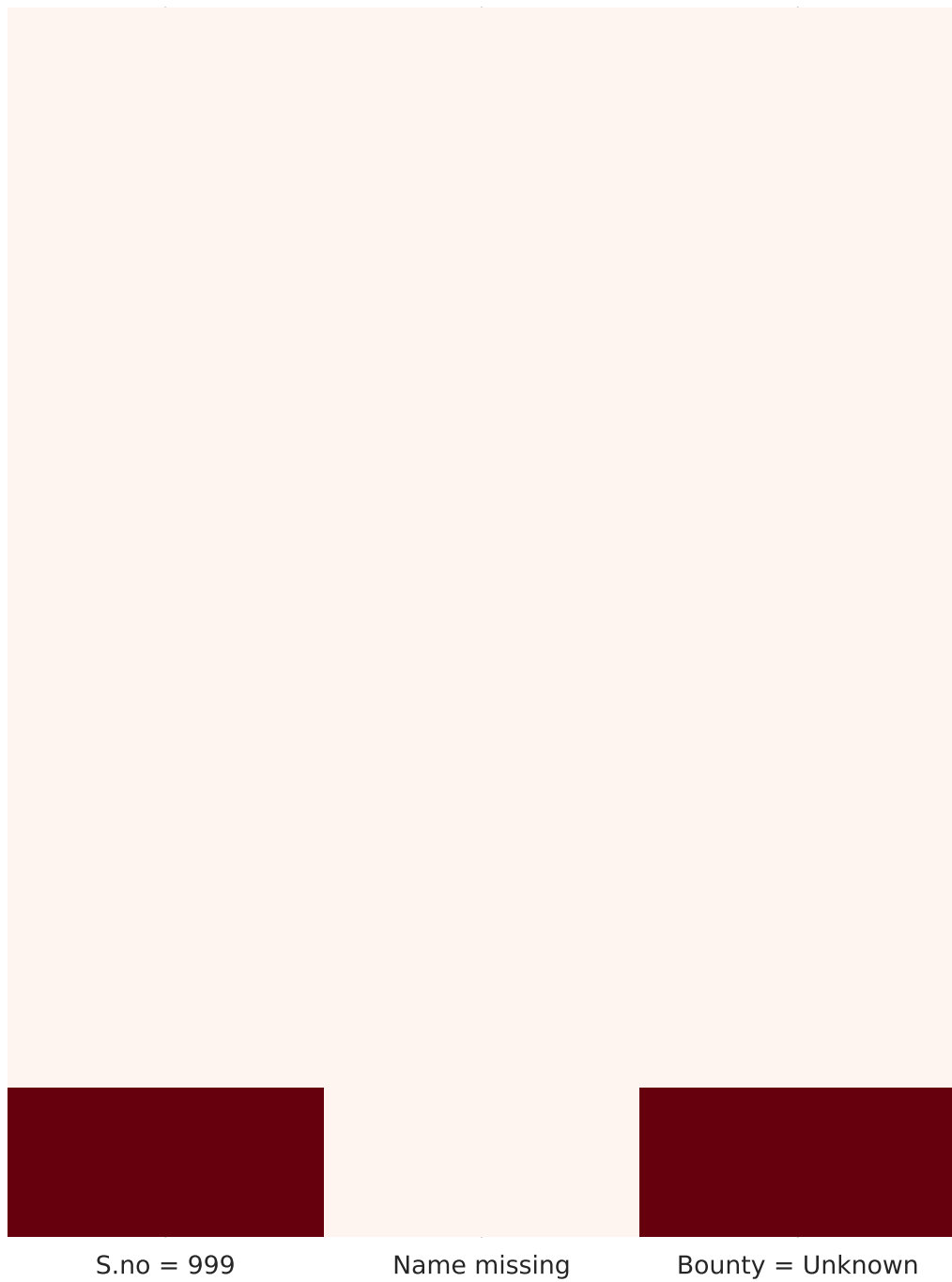
Raw shape : (165, 3)  
Clean shape : (165, 8)  
Known bounties : 145  
Unknown bounties : 20

#### Issues resolved

- Headers standardised: 3
- Names tidied: 0
- Duplicate pirates removed: 0
- Bounty strings parsed to numbers: 145
- Wrong currency symbols fixed: 1
- Unknown bounties flagged (not imputed): 20
- Placeholder ranks (999) nulled: 20
- Rank mismatches rebuilt: 0
- Outliers flagged (log IQR): 2

Skewness before log : 4.28  
Skewness after log : -2.44  
After log, minus outliers : 0.03

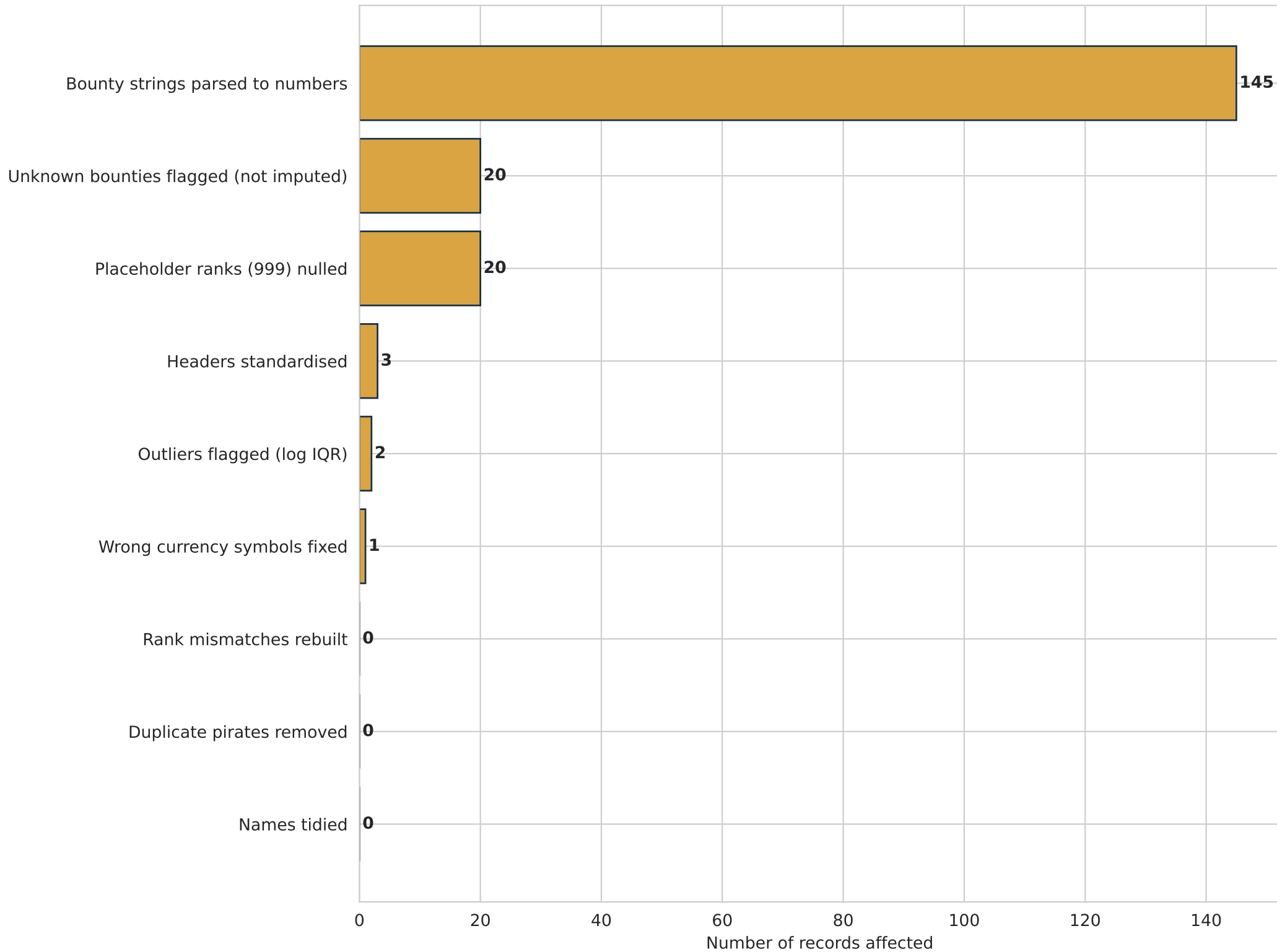
RAW: hidden missing values



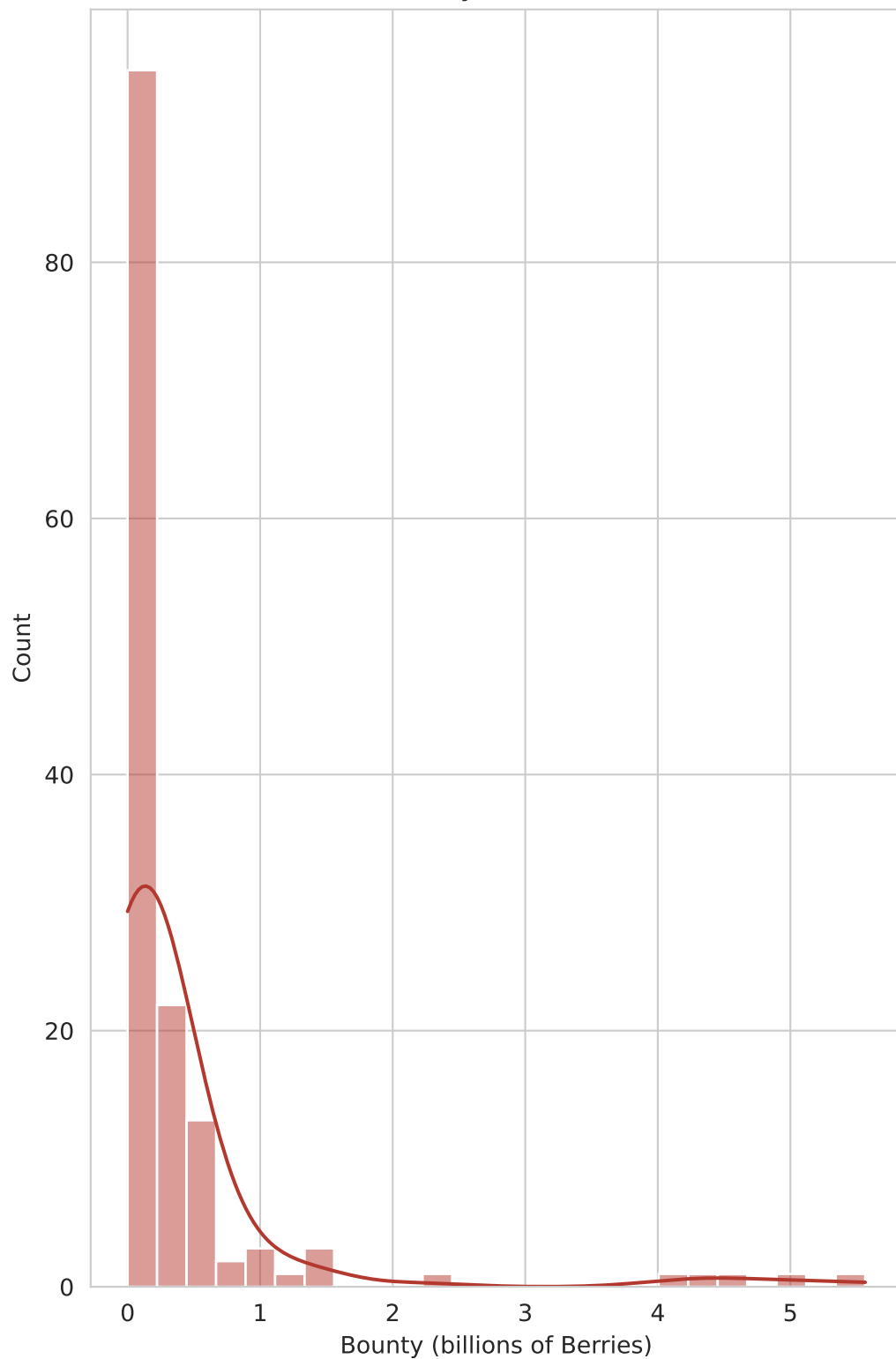
CLEAN: explicit NaN + flags



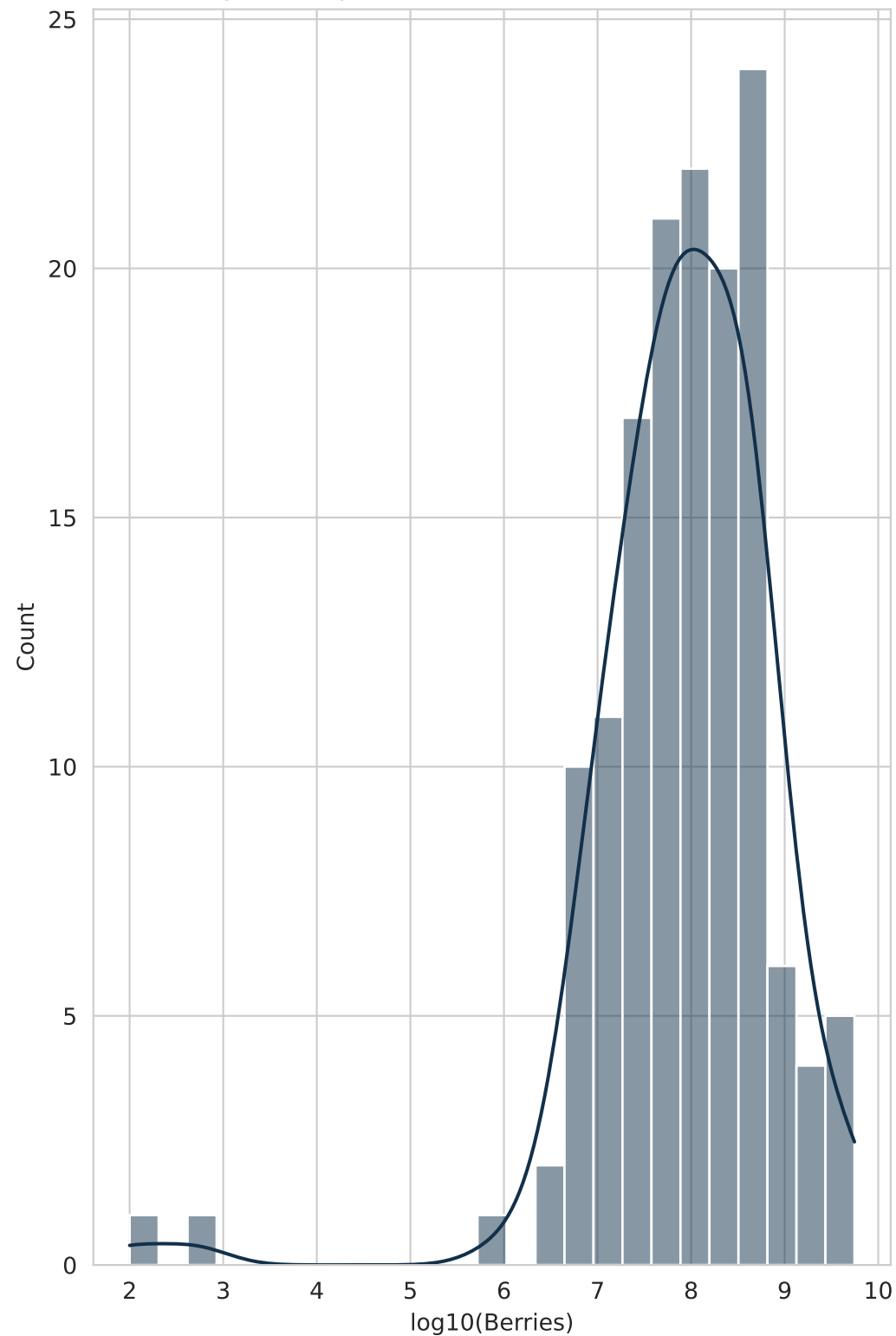
Data issues resolved at each pipeline step



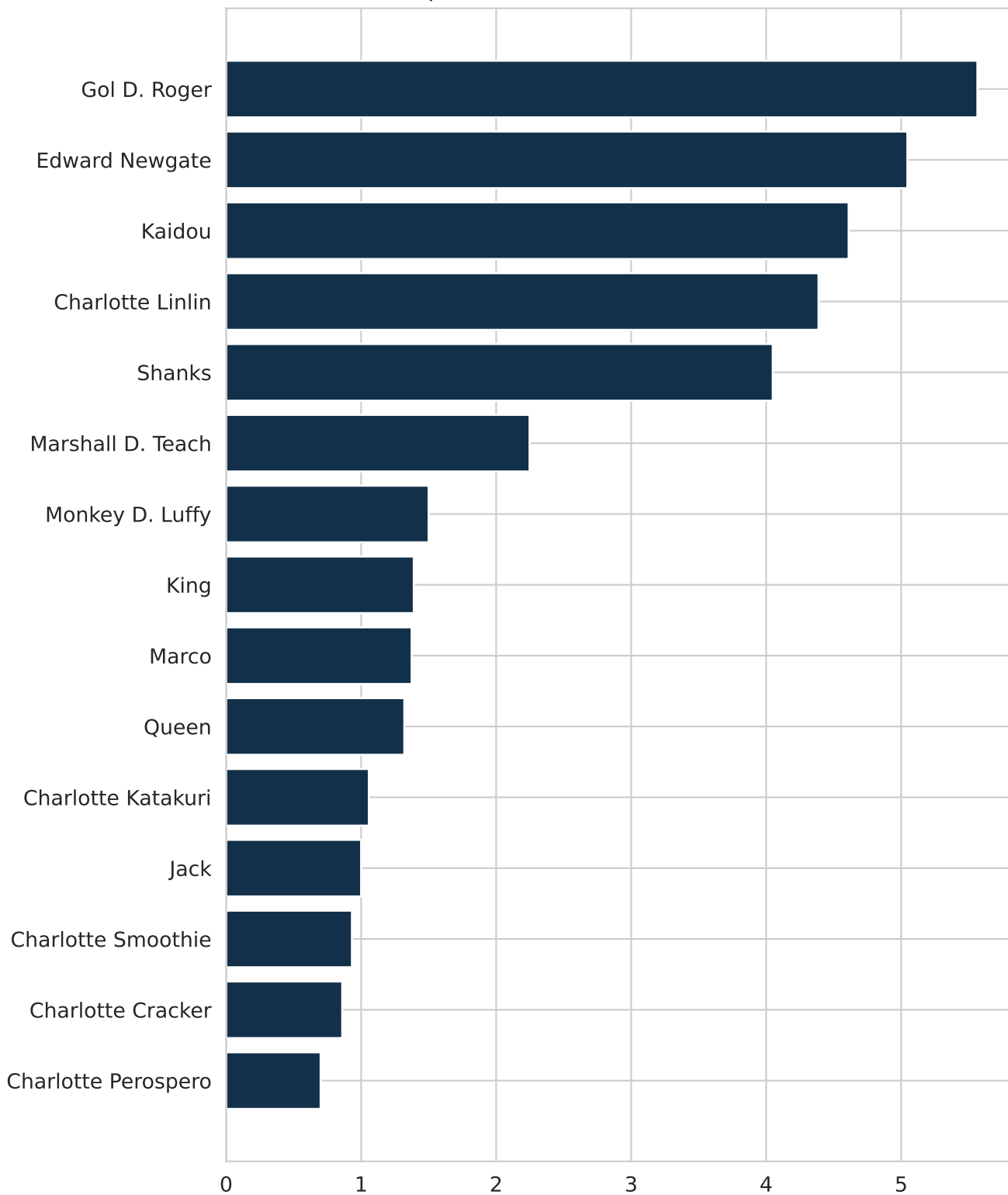
Raw bounty (skew = 4.28)



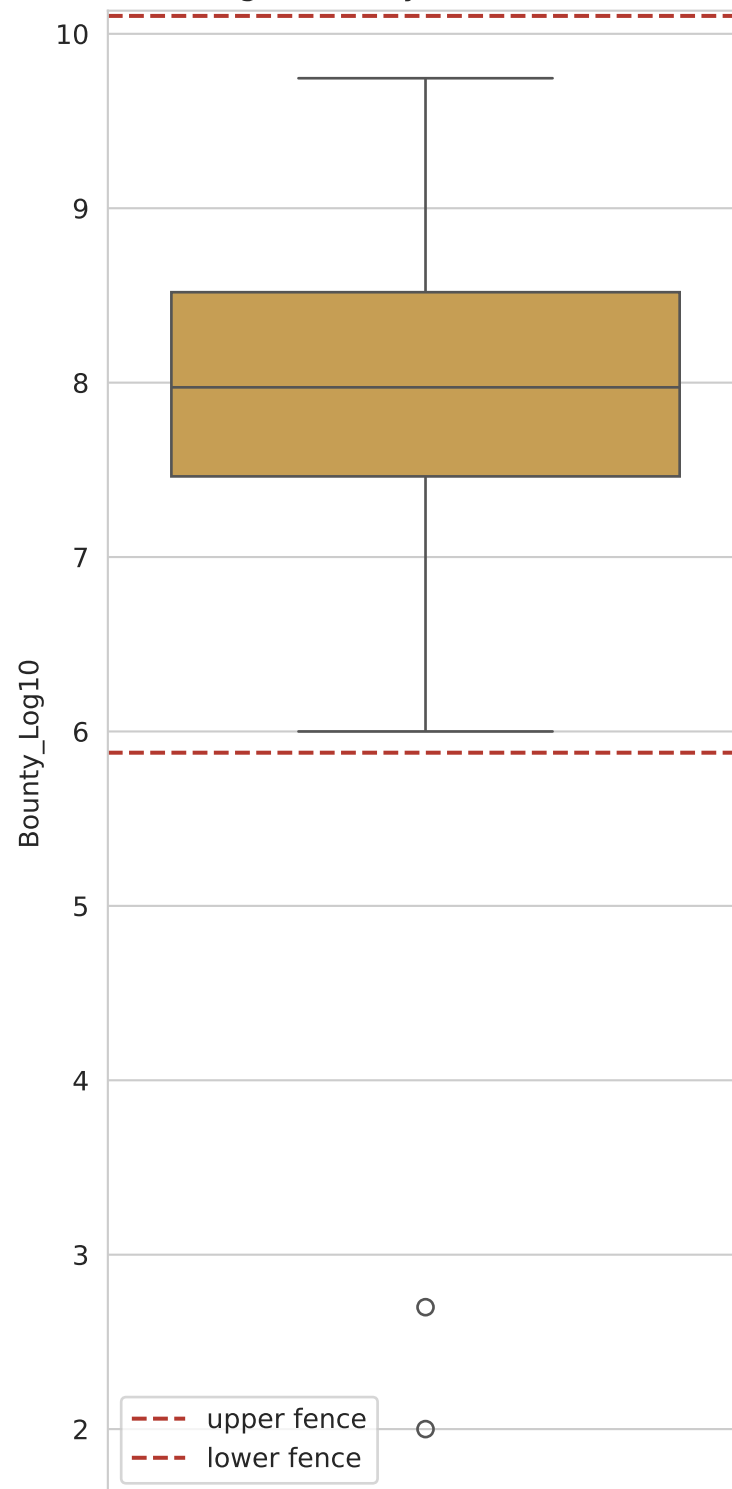
log10 bounty (skew = -2.44; 0.03 without outliers)



Top 15 bounties (billions of Berries)



log10 bounty + IQR fences



## PIPELINE AUDIT TRAIL

|                            |          |          |
|----------------------------|----------|----------|
| 01. Load raw data          | rows=165 | nulls=0  |
| 02. Standardise headers    | rows=165 | nulls=0  |
| 03. Clean text fields      | rows=165 | nulls=0  |
| 04. Remove duplicate names | rows=165 | nulls=0  |
| 05. Parse bounty strings   | rows=165 | nulls=20 |
| 06. Currency sanity check  | rows=165 | nulls=20 |
| 07. Flag unknown bounties  | rows=165 | nulls=20 |
| 08. Rebuild rank           | rows=165 | nulls=40 |
| 09. Sort + scale bounty    | rows=165 | nulls=80 |
| 10. Flag outliers          | rows=165 | nulls=80 |
| 11. Validation passed      | rows=165 | nulls=80 |